

CSSE 220 – Object-Oriented Software Development
Rose-Hulman Institute of Technology

Worksheet 05

Name (Print): _____ Section: _____

1. **Actual Value vs Address**

1) primitive variables store _____

2) Object variables store _____

2. Draw a diagram for:

int x = 10; int y = 20;

3. Draw a diagram showing what this creates in memory (include variable and object):

Player p1 = new Player(10);

4. **Value vs Reference**

1) When a primitive variable is assigned to another, primitive creates a copy of _____

2) Modifications to one variable (Do / Do Not) _____ affect the other variable

5. What are the final values of x,y,z:

Answer: _____

```
1 int x = 5;  
2 int y = x;  
3 int z = y;  
4 x = 15;  
5 z = 20;
```

6. 1) When an object is assigned to another, it creates a copy of _____
- 2) Modifications to one variable _____ the other variable
7. What is the value of p2.health?

Answer: _____

```
1 Player p1 = new Player(100);  
2 Player p2 = p1;  
3 p1.health = 50;
```

8. Draw boxes for:

```
int x = 10;  
int y = x;  
y = 20;
```

9. Equality

- 1) `==` compares _____
- 2) `.equals()` compares _____
10. True/False Strings are mutable (=can be modified once created)

11. Consider this code:

```
1 String s = "diamond";  
2 s = s + " ring";
```

Answer the following questions:

- 1) What is the value of s after execution? _____
- 2) Is the original "diamond" string modified? _____
- 3) How many String objects exist after this code runs? _____
12. **Before You Leave**
- Write one question you still have about classes, objects, or constructors.